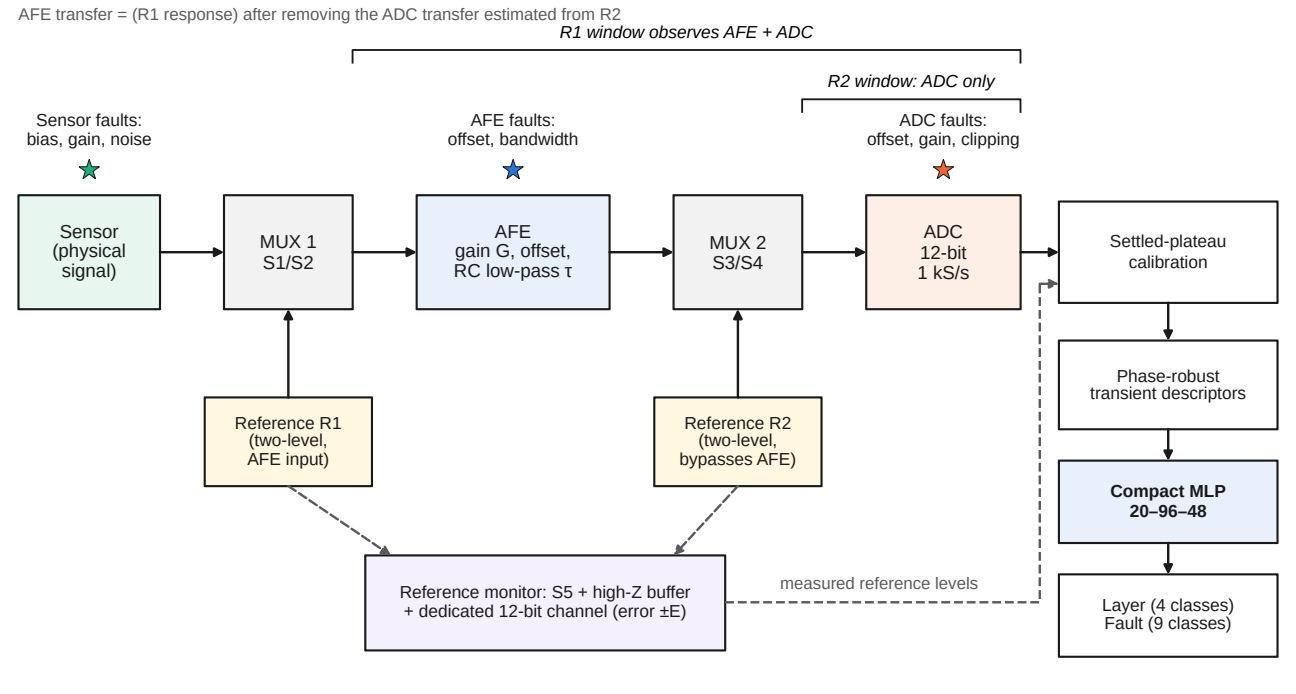
AFE transfer = (R1 response) after removing the ADC transfer estimated from R2
R1 window observes AFE + ADC
R2 window: ADC only
Sensor faults:
bias, gain, noise
AFE faults:
offset, bandwidth
ADC faults:
offset, gain, clipping
Sensor
(physical
signal)
MUX 1
S1/S2
AFE
gain G, offset,
RC low-pass τ
MUX 2
S3/S4
ADC
12-bit
1 kS/s
Settled-plateau
calibration
Phase-robust
transient descriptors
Compact MLP
20–96–48
Layer (4 classes)
Fault (9 classes)
Reference R1
(two-level,
AFE input)
Reference R2
(two-level,
bypasses AFE)
Reference monitor: S5 + high-Z buffer
+ dedicated 12-bit channel (error ±E)
measured reference levels